Chris (Shaopeng) Lin

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Education

Ph.D. Student in Computer Science — Toronto, Canada
University of Toronto — *Sep. 2024 – Present*

Bachelor of Science: Computer Science — Scarborough, Canada
University of Toronto Scarborough — *Sept. 2019 – Sept. 2024*

- **CGPA**: 3.97/4.0
- **Dean's List**: Fall 2020 - Fall 2024
- **Awards**: UTEA Summer 2024 - CAD 7500

Publications

PREPRINTS AND IN SUBMISSION

PrisonBreak: Jailbreaking Large Language Models with at Most Twenty-Five Targeted Bit-flips
Zachary Coalson, Jeonghyun Woo, Chris S. Lin, Joyce Qu, Yu Sun, Shiyang Chen, Lishan Yang, Gururaj Saileshwar, Prashant Nair, Bo Fang, Sanghyun Hong
In submission, *The Fortieth Annual Conference on Neural Information Processing Systems*, 2026. (**NeurIPS '26**)

CONFERENCE PUBLICATIONS

GPUThor: Amplifying Rowhammer Attacks via Non-Uniform Patterns to Exploit ECC-Protected GPUs
Chris S. Lin, Joyce Qu, Aditya Rajeev, Gururaj Saileshwar
In *the 2026 ACM SIGSAC Conference on Computer and Communications Security (**CCS '26**)*

- **Media Coverage:** BleepingComputer, The Hacker News, CSO Online, TechRadar.

GPUBreach: Privilege Escalation Attacks on GPUs using Rowhammer
Chris S. Lin, Yuqin Yan, Guozhen Ding, Joyce Qu, Joseph Zhu, David Lie, Gururaj Saileshwar
In *47th IEEE Symposium on Security and Privacy*, 2026. (**S&P '26**) and Black Hat 2026

- **Media Coverage:** Ars Technica, BleepingComputer, The Hacker News.

🏆 **S&P Distinguished Paper Award**
🏆 **Google Bug Bounty Award - USD 600**

GPUHammer: Rowhammer Attacks on GPU Memories are Practical
Chris S. Lin, Joyce Qu, Gururaj Saileshwar
In *34th USENIX Security Symposium*, August 2025. (**SEC'25**)

- **Media Coverage:** Ars Technica, Tom's Hardware, The Register, BleepingComputer.

🏆 **CSAW'25 (North America) Best Paper Award**
🏆 **IEEE Micro Top Picks 2026 - Honorable Mention**

QPRAC: Towards secure and practical prac-based rowhammer mitigation using priority queues
Jeonghyun Woo, Chris S. Lin, Prashant J. Nair, Aamer Jaleel, Gururaj Saileshwar
IEEE International Symposium on High Performance Computer Architecture, 2025. (**HPCA '25**)

🏆 **Distinguished Artifact Award**

Experience

Graduate Research Assistant — Jan. 2024 - Present
University of Toronto — *Toronto, Canada*

- Designed and evaluated novel Rowhammer attack patterns on the NVIDIA A6000 GPU.
- Developed and demonstrated new GPU-based exploit techniques, including LLM jailbreaking and GPU privilege-escalation, improving understanding of emerging accelerator-centric security risks.
- Investigated and prototyped compiler-assisted GPU taint tracking mechanisms to enforce multi-user data isolation on batched ML inference.

Open Source Developer: Google Summer of Code May. 2024 - Sep. 2024
Kiwix *Remote*

- Resolved 17 long-standing issues in the Kiwix-Desktop repository, leading to the successful completion of a 350+ hour GSOC project.
- Addressed two major issues persisting for 4+ years: (1) implemented a table of contents for Kiwix-Desktop web pages using the Javascript-QT Webchannel API, and (2) enhanced search suggestion display by refactoring the QT MVC framework.

GPU Compiler Engineer Co-op Sep. 2023 - Dec. 2023
Huawei Technologies Co., Ltd. *Markham, Canada*

- Achieved 50+% decrease in encoding time by refactoring 200+ lines of 3 major instruction sets lowering code from run-time to compile-time through auto-generated C++ by leveraging LLVM TableGen.
- Researched and presented to the team a solution to incorporate Inline Assembly into our backend optimization pipeline, which is now a part of the team milestone.

Software Developer Co-op May. 2022 - Sep. 2022
Safe Software Inc. *Surrey, Canada*

- Lowered client misuse of the FME language IDE by 60%, by developing a self-proposed core error-logging and notification feature.
- Led development for 1/3 of the translation GUI in the JSON to FME's domain-specific language IDE, leveraging the QT MVC framework on Windows.

Software Developer Co-op May. 2021 - Dec. 2021
DataOcean AI *Beijing, China*

- Developed an NLP regex transcription tagging module with 97% accuracy, by collaborating with Spanish experts to produce the pipeline with C++ and Python.
- Led a code redesign achieving a decrease of 60% in compilation time and 30% in storage space on mobile devices, by merging multiple language translation executables into a single instance.

Teaching

Teaching Assistant Sep. 2022 - Present
University of Toronto

- Experienced in delivering lectures and interactive tutorials to groups ranging from approximately 30-300 students.
- Consistently recognized in student feedback for conducting engaging tutorials and being a valuable member of the teaching team.
- Courses: Computer and Network Security, Operating Systems, Computer Organization, Introduction to Computer Science, Discrete Mathematics

Technical Skills

Languages: C/C++, CUDA, Python, Assembly, Bash, JavaScript, HTML/CSS, SQL, Java, Kotlin
Frameworks: Qt, LLVM, React, Node.js, NextJS, Google Test
Developer Tools: Linux, Git, Docker, CMake, MongoDB, Postgres, Redis
Misc: Object Oriented Programming, TCP/IP, Multi-threading, Performance Benchmarking
Relevant Courses: Compiler Optimization, Computer Architecture, Neural Networks and Deep Learning

Awards & Recognitions

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🏆**IEEE Micro Top Picks 2026 - Honorable Mention**

🏆**CSAW 2025 (North America) Best Paper Award**

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